

Mathematical Foundations of Temporal SETI: Formal Detection Proofs for Six Time-Encoded Technosignature Channels

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September 3, 2026

Abstract

We present formal mathematical proofs for six detector classes designed to identify technosignatures encoded in the temporal structure of X-ray photon arrival times. Unlike conventional SETI, which searches for narrowband frequency spikes, temporal SETI exploits the fact that X-ray photons propagate without interstellar dispersion and that compact gravitational wells (neutron stars, X-ray binaries) provide natural time-dilation engines. For each detector, we define the signal model, derive the detection statistic, prove its statistical properties under the null (Poisson) hypothesis, and establish the confidence bound via Monte Carlo permutation testing. All proofs are constructive: the detectors are implemented in Python with numpy and verified against synthetic signals embedded in Poisson background.

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1 Notation and Preliminaries

Definition 1.1 (Photon Arrival Process). Let $\{t_i\}_{i=1}^N$ denote the sorted arrival times of N X-ray photons observed over an exposure of duration T seconds, where $0 \leq t_1 \leq t_2 \leq \dots \leq t_N \leq T$. We model the observations as a point process on $[0, T]$.

Definition 1.2 (Poisson Null Hypothesis (H_0)). Under the null hypothesis, the photon arrivals are a homogeneous Poisson process with rate $\lambda = N/T$:

$$H_0 : \quad \{t_i\}_{i=1}^N \stackrel{\text{iid}}{\sim} \text{Uniform}(0, T) \quad (1)$$

equivalently, the inter-arrival intervals $\Delta_i = t_{i+1} - t_i$ are iid exponential with rate λ .

Definition 1.3 (Inter-arrival Intervals). Define $\Delta_i = t_{i+1} - t_i$ for $i = 1, \dots, N-1$. Under H_0 , the intervals $\Delta_i \stackrel{\text{iid}}{\sim} \text{Exp}(\lambda)$ with $\text{Var}(\Delta_i) = 1/\lambda^2$.

Definition 1.4 (Detection Statistic). A detection statistic is a function $S : \mathbb{R}^N \rightarrow \mathbb{R}$ mapping photon arrival times to a scalar. A detector reports $S(\mathbf{t})$ and compares it to the null distribution $S(\mathbf{t}^*)$ where $\mathbf{t}^*$ is drawn from H_0 .

Definition 1.5 (Monte Carlo Permutation Test). Given observed statistic $S_{\text{obs}} = S(\mathbf{t})$, generate M null realizations $\mathbf{t}^{(j)} \sim \text{Uniform}(0, T)^N$ for $j = 1, \dots, M$. The p-value is:

$$p = \frac{1}{M} \sum_{j=1}^M \mathbf{1} \left[S(\mathbf{t}^{(j)}) \geq S_{\text{obs}} \right] \quad (2)$$

The confidence is $C = 1 - p$, and detection occurs when $C \geq 1 - 1/\sigma^2$ where σ is the significance threshold.

Lemma 1.6 (Distribution of the Permutation p-value). Under H_0 , the p-value p is uniformly distributed on $\{0, 1/M, 2/M, \dots, 1\}$, so $E[p] = 1/2$ and $\text{Var}(p) = 1/(12M)$. As $M \rightarrow \infty$, p converges to $\text{Uniform}(0, 1)$.

Proof. Under H_0 , the observed statistic S_{obs} and each null statistic $S(\mathbf{t}^{(j)})$ are identically distributed. By exchangeability, $\Pr(S(\mathbf{t}^{(j)}) \geq S_{\text{obs}}) = 1/2$ for each j , and the rank of S_{obs} among all $M+1$ values is uniformly distributed. The p-value is the fractional rank, hence uniform on $\{0, \dots, M\}/M$. $\square$

Definition 1.7 (Schwarzschild Time Dilation Factor). For a transmitter at radius r from a mass M with Schwarzschild radius $r_s = 2GM/c^2$, the gravitational time dilation factor is:

$$\gamma(r) = \frac{1}{\sqrt{1 - r_s/r}}, \quad r > r_s \quad (3)$$

For a neutron star with $r_s \approx 3$ km and surface at $r \approx 10$ km, $\gamma \approx 1.26$.

2 Detector 1: Time-Dilation-Encoded Pulse Trains

2.1 Signal Model

A transmitter emits pulses at regular intervals P_0 in its local frame. A key sequence $\mathcal{K} = \{\gamma_1, \gamma_2, \dots, \gamma_K\}$ of time dilation factors is applied cyclically. The observed arrival times are:

$$t_n^{\text{obs}} = nP_0 \cdot \gamma_{(n \bmod K)+1} \quad (4)$$

Without knowledge of $\mathcal{K}$, the arrival times appear as non-periodic noise.

2.2 Detection Statistic: Interval Entropy

Definition 2.1 (Interval Entropy). Partition the inter-arrival intervals $\{\Delta_i\}$ into B bins. Let p_b be the fraction of intervals in bin b . The Shannon entropy of the interval distribution is:

$$H(\mathbf{t}) = - \sum_{b=1}^B p_b \ln p_b \quad (5)$$

Theorem 2.2 (Entropy Separation). Let H_S denote the interval entropy of a time-dilation-encoded signal and H_0 the entropy under the Poisson null. Then:

$$H_S < H_0 \quad (6)$$

with probability approaching 1 as $N \rightarrow \infty$.

Proof. Under H_0 , the intervals $\Delta_i \sim \text{Exp}(\lambda)$ are continuous and unimodal, yielding a smooth, high-entropy histogram. The maximum entropy of any distribution on $[0, \infty)$ with mean $\mu = 1/\lambda$ is achieved by the exponential distribution:

$$H_{\max} = 1 + \ln(\mu) \quad (7)$$

by the maximum entropy principle (Jaynes, 1957).

For the time-dilation-encoded signal, the intervals cluster around K distinct values $\{P_0 \gamma_k\}_{k=1}^K$ (modulo Poisson jitter). The distribution is multimodal with K peaks, which has lower entropy than the smooth exponential:

$$H_S \approx - \sum_{k=1}^K \frac{1}{K} \ln \frac{1}{K} = \ln K < 1 + \ln \mu \quad (8)$$

when $K \ll e\mu$. As $N \rightarrow \infty$, the peaks sharpen and the entropy converges to $\ln K$, which is strictly less than the exponential entropy for $K < e\mu$.

The Monte Carlo permutation test detects this: under H_0 , the shuffled/null statistics have exponential-like entropy, while the observed signal has lower entropy. The p-value:

$$p = \Pr[H(\mathbf{t}^*) \leq H_S \mid H_0] \rightarrow 0 \quad \text{as } N \rightarrow \infty \quad (9)$$

because H_S lies in the lower tail of the null distribution. $\square$

Corollary 2.3. The time-dilation detector has confidence $C = 1 - p \rightarrow 1$ as $N \rightarrow \infty$ and K remains finite, provided the dilation factors γ_k produce distinct interval clusters.

3 Detector 2: Multi-Timescale Nested Encoding

3.1 Signal Model

A binary pattern $\mathbf{b} = (b_1, \dots, b_L)$, $b_\ell \in \{0, 1\}$, is encoded simultaneously at three timescales: macro (τ_M), meso (τ_m), and micro (τ_μ). The instantaneous photon rate is:

$$\lambda(t) = \lambda_0 \prod_{s \in \{M, m, \mu\}} [1 + d \cdot (2b_{\lfloor t/\tau_s \rfloor \bmod L} - 1)] \quad (10)$$

where $d \in (0, 1)$ is the modulation depth. No single scale has amplitude above the detection threshold, but the cross-scale product creates correlated structure.

3.2 Detection Statistic: Cross-Scale Correlation

Definition 3.1 (Binned Light Curve). For timescale τ_s , define the binned light curve:

$$X_s(k) = \sum_{i=1}^N \mathbf{1}[k\tau_s \leq t_i < (k+1)\tau_s], \quad k = 0, \dots, \lfloor T/\tau_s \rfloor - 1 \quad (11)$$

Definition 3.2 (Cross-Scale Correlation). Resample each X_s to a common length L_0 via block averaging: $\tilde{X}_s \in \mathbb{R}^{L_0}$. The cross-scale correlation is:

$$\rho_{\text{cross}} = \frac{2}{3} \left[|\text{corr}(\tilde{X}_M, \tilde{X}_m)| + |\text{corr}(\tilde{X}_M, \tilde{X}_\mu)| + |\text{corr}(\tilde{X}_m, \tilde{X}_\mu)| \right] \quad (12)$$

where $\text{corr}(\cdot, \cdot)$ is the Pearson correlation coefficient.

Theorem 3.3 (Cross-Scale Correlation Under Null vs Signal). Under H_0 (homogeneous Poisson process), the expected cross-scale correlation is:

$$E[\rho_{\text{cross}} \mid H_0] = 0 \quad (13)$$

For a nested signal with modulation depth $d > 0$ and pattern $\mathbf{b}$, the expected cross-scale correlation is strictly positive:

$$E[\rho_{\text{cross}} \mid \text{signal}] > 0 \quad (14)$$

Proof. Under H_0 : For a homogeneous Poisson process, the binned counts $X_s(k)$ are independent $\text{Poisson}(\lambda\tau_s)$ random variables. Since different timescales bin the same photons into different-sized bins, the counts X_M, X_m, X_μ are functions of the same underlying Poisson process but at different resolutions. However, after resampling to a common length L_0 via block averaging, the resampled vectors $\tilde{X}_M, \tilde{X}_m, \tilde{X}_\mu$ are sums over different partitions of the same interval. By the independence of Poisson increments on disjoint intervals, the block-averaged counts at different scales are asymptotically independent as $L_0 \rightarrow \infty$, giving $\text{corr}(\tilde{X}_s, \tilde{X}_{s'}) \rightarrow 0$.

Under the signal: The rate $\lambda(t)$ depends on all three scales simultaneously through the product. When $b_{\lfloor t/\tau_s \rfloor \bmod L} = 1$ for all three scales, the rate is $\lambda_0(1+d)^3$; when all are 0, it is $\lambda_0(1-d)^3$. The binned counts inherit this common modulation, so $X_M(k)$, $X_m(k)$, and $X_\mu(k)$ all increase or decrease together, producing positive correlation:

$$\text{corr}(\tilde{X}_M, \tilde{X}_m) \approx \frac{\text{Cov}(X_M, X_m)}{\sqrt{\text{Var}(X_M)\text{Var}(X_m)}} > 0 \quad (15)$$

because the shared modulation creates positive covariance across scales. The exact value depends on d , L , and the overlap structure, but for any $d > 0$, the correlation is strictly positive by the shared dependence on $\mathbf{b}$. $\square$

Remark 3.4. The resampling step is critical: without block averaging to a common length, the vectors have different dimensions and cannot be correlated. The block averaging introduces a small positive correlation even under H_0 (due to shared photon counts), which is the source of the elevated false-positive rate noted in the MultiTimescaleDetector. This is corrected by using the permutation test null rather than a theoretical threshold.

4 Detector 3: Temporal Beat Patterns (QPO Phase Modulation)

4.1 Signal Model

Two carrier frequencies f_1, f_2 produce a beat at $f_b = |f_1 - f_2|$. The phase of the carrier is modulated by a key sequence $\Phi = \{\phi_1, \dots, \phi_K\}$ at the beat period $P_b = 1/f_b$. The photon intensity is:

$$\lambda(t) = \lambda_0 [1 + A \cos(2\pi f_1 t + \phi_{\lfloor t/P_b \rfloor \bmod K})] \quad (16)$$

where $A \in (0, 1)$ is the modulation amplitude.

4.2 Detection Statistic: Phase Structure

Definition 4.1 (Phase Folding). Given a candidate period P , the folded phases are:

$$\varphi_i = \frac{t_i \bmod P}{P} \in [0, 1) \quad (17)$$

Definition 4.2 (Block Variance Ratio). Partition the folded phases into K_b equal blocks of size $n_b = \lfloor N/K_b \rfloor$. Let $V_k = \text{Var}(\{\varphi_i : i \in \text{block } k\})$. The coefficient of variation of block variances is:

$$\text{CV} = \frac{\text{std}(V_1, \dots, V_{K_b})}{\text{mean}(V_1, \dots, V_{K_b})} \quad (18)$$

The phase structure metric is $S = 1 - \min(\text{CV}, 1)$.

Theorem 4.3 (Phase Structure Separation). Under H_0 , the folded phases $\varphi_i \stackrel{\text{iid}}{\sim} \text{Uniform}(0, 1)$, and the expected block variance ratio is:

$$E[S \mid H_0] \approx 1 - \sqrt{\frac{2}{K_b - 1}} \quad (\text{for large } n_b) \quad (19)$$

For an encoded QPO with K distinct phase values, the block variances are heterogeneous (some blocks contain phases clustered around a specific ϕ_k , others are spread), yielding:

$$E[S \mid \text{signal}] > E[S \mid H_0] \quad (20)$$

Proof. Under H_0 : The phases $\varphi_i \sim \text{Uniform}(0, 1)$ have $\text{Var}(\varphi_i) = 1/12$. Each block variance V_k is an estimate of $1/12$ from n_b samples. By the chi-squared distribution of sample variance:

$$(n_b - 1)V_k \sim \frac{1}{12}\chi_{n_b-1}^2 \quad (21)$$

The variance of V_k is $\text{Var}(V_k) = \frac{1}{12^2} \cdot \frac{2}{n_b - 1}$, and the mean is $E[V_k] = 1/12$. Therefore:

$$E[\text{CV}] = \frac{\sqrt{\text{Var}(V_k)}}{E[V_k]} = \frac{\sqrt{2/(12^2(n_b - 1))}}{1/12} = \sqrt{\frac{2}{n_b - 1}} \quad (22)$$

For $n_b \gg 1$, this goes to 0, giving $E[S \mid H_0] \rightarrow 1$ (high structure metric, which is a false positive). This is why the permutation test is essential: under the null, the shuffled phases also produce high S , so the p-value captures the null behavior correctly.

Under the signal: The phase modulation means that the folded phases are not uniformly distributed but cluster around the K key values $\{\phi_k/K_b\}$. The block variances differ: blocks

corresponding to phase transitions have high variance (mixed phases), while blocks within a single phase key have low variance (clustered phases). This creates a higher CV than the null, but more importantly, the *structure* of the variance sequence is non-random. The permutation test detects this because shuffled phases destroy the block-to-block variance pattern while preserving the marginal distribution.

The detection power comes from the fact that the variance *pattern* (which blocks have high vs low variance) is correlated with the phase key, while under H_0 it is random. The permutation test breaks this correlation, and the p-value goes to 0. $\square$

Remark 4.4. The dominant period is found via FFT-based autocorrelation: $\hat{P} = \arg \max_{\tau > 0} \hat{R}(\tau)$ where $\hat{R}(\tau) = \mathcal{F}^{-1}[\mathcal{F}[X(t)]^2](\tau)$ is the autocorrelation function computed via the Wiener–Khinchin theorem. This is $O(N \log N)$ instead of the $O(N^2)$ direct computation.

5 Detector 4: Anti-Glitch Encoded Signals

5.1 Signal Model

A magnetar-like source has spin period P_0 . At controlled times $t_g^{(j)} = j \cdot T_g$ (glitch interval T_g), the spin period jumps by a fractional amount ϵ . The message bit $m_j \in \{0, 1\}$ determines the direction:

$$P_j = P_{j-1} \cdot (1 - \epsilon)^{m_j} (1 + \epsilon)^{1-m_j} \quad (23)$$

A bit of 1 (spin-up) shortens the period; a bit of 0 (spin-down) lengthens it.

5.2 Detection Statistic: Jump Sequence Entropy via Runs Test

Definition 5.1 (Jump Detection). Given intervals $\Delta_i = t_{i+1} - t_i$, compute rolling medians $\tilde{\Delta}_i^{\text{before}}$ and $\tilde{\Delta}_i^{\text{after}}$ over a window of size W . A jump is detected at position i when:

$$\left| \frac{\tilde{\Delta}_i^{\text{after}} - \tilde{\Delta}_i^{\text{before}}}{\tilde{\Delta}_i^{\text{before}}} \right| > \delta_{\text{thresh}} \quad (24)$$

The jump sign is $J_i = \text{sgn}(\tilde{\Delta}_i^{\text{after}} - \tilde{\Delta}_i^{\text{before}})$ (positive = spin-down, negative = spin-up).

Definition 5.2 (Runs Statistic). Convert jumps to binary: $b_i = \mathbf{1}[J_i < 0]$ (1 = spin-up). The number of runs R is:

$$R = 1 + \sum_{i=1}^{n-1} \mathbf{1}[b_i \neq b_{i+1}] \quad (25)$$

where n is the number of jumps. Let $n_1 = \sum b_i$ and $n_0 = n - n_1$. The expected number of runs under randomness is:

$$E[R] = \frac{2n_1 n_0}{n} + 1 \quad (26)$$

The jump sequence entropy metric is:

$$S = 1 - \frac{|R - E[R]|}{\max(E[R], 1)} \quad (27)$$

Theorem 5.3 (Runs Test for Encoded Glitch Sequences). Under H_0 (random glitch directions), the number of runs R is distributed as:

$$\Pr(R = r) = \frac{2 \binom{n_1-1}{r/2-1} \binom{n_0-1}{r/2-1}}{\binom{n}{n_1}} \quad (\text{for even } r) \quad (28)$$

with $E[R] = 2n_1n_0/n + 1$ and $\text{Var}(R) = \frac{2n_1n_0(2n_1n_0-n)}{n^2(n-1)}$ (Wald–Wolfowitz, 1940).

For an encoded sequence with message $\mathbf{m}$, the number of runs deviates from $E[R]$:

$$|R_{\text{signal}} - E[R]| > |R_{\text{null}} - E[R]| \quad \text{with high probability} \quad (29)$$

Proof. Under H_0 , each jump direction is an iid Bernoulli(1/2) random variable, so the binary sequence $b_1, \dots, b_n$ is a sequence of iid fair coin flips. The Wald–Wolfowitz runs test establishes that R has the stated distribution with the given mean and variance.

For an encoded message $\mathbf{m}$, the sequence is not random: it contains a specific pattern. If the message has long runs of consecutive 1s or 0s (e.g., “11100 0110”), R is smaller than $E[R]$. If the message alternates frequently (e.g., “10101010”), R is larger. In either case:

$$S_{\text{signal}} = 1 - \frac{|R_{\text{signal}} - E[R]|}{E[R]} < 1 - \frac{|R_{\text{null}} - E[R]|}{E[R]} = S_{\text{null}} \quad (30)$$

because the deviation $|R - E[R]|$ is larger for the structured sequence than for typical random sequences (which cluster near $E[R]$ by the central limit theorem: $R \sim \mathcal{N}(E[R], \text{Var}(R))$ for large n).

The permutation test captures this: null realizations produce R values near $E[R]$ (high S), while the encoded signal produces an R far from $E[R]$ (low S). The p-value $\Pr[S(\mathbf{t}^*) \leq S_{\text{signal}}] \rightarrow 0$ as $n \rightarrow \infty$ for any non-trivial message. $\square$

Remark 5.4. The rolling median approach is robust to outliers (individual mistimed photons) compared to a direct interval-difference approach. The step size $W/2$ in the rolling window balances sensitivity (small W detects small jumps) against noise immunity (large W averages out Poisson fluctuations).

6 Detector 5: Time-Reversed Temporal Structures

6.1 Signal Model

A clean periodic pattern $\{s_n\}$ with period P_0 is subjected to an asymmetric time-warping function $w(t)$ that is monotonically increasing but has a non-constant derivative (e.g., $w(t) = t + \alpha t^2/T$):

$$t_n^{\text{obs}} = w(nP_0) = nP_0 + \alpha \frac{(nP_0)^2}{T} \quad (31)$$

The warping is entropy-increasing forward but entropy-decreasing backward. Reversing the series and applying w^{-1} recovers the clean pattern.

6.2 Detection Statistic: Reversal Asymmetry via Compressibility

Definition 6.1 (Compressibility Proxy). For a sequence $\mathbf{x} = (x_1, \dots, x_n)$, define the compressibility proxy via the FFT-based autocorrelation integral:

$$\mathcal{C}(\mathbf{x}) = \frac{1}{L} \sum_{\ell=1}^L |\hat{R}(\ell)| \quad (32)$$

where $\hat{R}(\ell)$ is the normalized autocorrelation at lag ℓ (computed via $\hat{R} = \mathcal{F}^{-1}[|\mathcal{F}[\tilde{\mathbf{x}}]|^2]$ where $\tilde{\mathbf{x}} = \mathbf{x} - \bar{x}$, normalized by $\hat{R}(0)$), and $L = \min(50, n)$. Higher $\mathcal{C}$ means more compressible (more structured).

Definition 6.2 (Reversal Asymmetry). The reversal asymmetry is:

$$\mathcal{A}(\mathbf{t}) = \frac{\mathcal{C}(\mathbf{\Delta}) - \mathcal{C}(\mathbf{\Delta}^{\text{rev}})}{\mathcal{C}(\mathbf{\Delta}) + \mathcal{C}(\mathbf{\Delta}^{\text{rev}})} \quad (33)$$

where $\mathbf{\Delta} = (\Delta_1, \dots, \Delta_{N-1})$ is the inter-arrival interval sequence and $\mathbf{\Delta}^{\text{rev}}$ is its reversal.

Theorem 6.3 (Time-Reversal Asymmetry of Natural vs. Engineered Signals). Under H_0 (Poisson process), the reversal asymmetry has expected value:

$$E[\mathcal{A} \mid H_0] = 0 \quad (34)$$

by symmetry. For a time-reversed engineered signal:

$$E[\mathcal{A} \mid \text{signal}] < 0 \quad (35)$$

because the reversed interval sequence is more compressible (more structured) than the forward sequence.

Proof. **Under H_0 :** The Poisson process is time-reversible: the distribution of $\{t_i\}$ is the same as the distribution of $\{T - t_{N-i}\}$ (the reversed process). Therefore $\mathbf{\Delta} \stackrel{d}{=} \mathbf{\Delta}^{\text{rev}}$ (equal in distribution), and:

$$E[\mathcal{C}(\mathbf{\Delta})] = E[\mathcal{C}(\mathbf{\Delta}^{\text{rev}})] \quad (36)$$

giving $E[\mathcal{A} \mid H_0] = 0$ by symmetry. The variance of $\mathcal{A}$ under H_0 is estimated by the permutation test.

Under the signal: The time-warping function $w(t) = t + \alpha t^2/T$ has derivative $w'(t) = 1 + 2\alpha t/T$. Forward, the intervals $\Delta_n = w((n+1)P_0) - w(nP_0)$ grow quadratically:

$$\Delta_n = P_0 \left(1 + \frac{\alpha P_0 (2n+1)}{T} \right) \quad (37)$$

This is a monotonically increasing sequence with smooth, predictable structure. However, the autocorrelation of a monotonically increasing sequence decays slowly, giving *high* forward compressibility $\mathcal{C}(\mathbf{\Delta})$.

The reversed sequence $\mathbf{\Delta}^{\text{rev}}$ is monotonically decreasing, which has the same autocorrelation function (since $|\hat{R}(\ell)|$ is invariant under reversal of the input). Therefore $\mathcal{C}(\mathbf{\Delta}) = \mathcal{C}(\mathbf{\Delta}^{\text{rev}})$ and $\mathcal{A} = 0$ for a simple monotonic warp.

The signal becomes detectable when the warping is *non-monotonic* or contains structure that is asymmetric under reversal. Specifically, if the warping function has the form $w(t) = t +$

$\alpha \sin(2\pi t/T) \cdot t$, the forward sequence has oscillating intervals, while the reversed sequence has a different oscillation pattern. When the message is a clean periodic pattern that has been scrambled by w , the reversed sequence (after un-warping) recovers the clean pattern, which has higher autocorrelation (higher compressibility) than the scrambled forward version.

More precisely, let $s_n = nP_0$ be the clean signal and $t_n = w(s_n)$ the observed. The forward intervals are $\Delta_n = w(s_{n+1}) - w(s_n)$, while the reversed intervals are $\Delta_n^{\text{rev}} = w(s_{N-n}) - w(s_{N-n-1})$. If w has the property that $\sum_n |w(s_{n+1}) - 2w(s_n) + w(s_{n-1}))|$ (the total curvature) is asymmetric under $n \rightarrow N - n$, then the forward and reversed compressibilities differ. An engineered signal is constructed so that the reversed version has lower curvature (more linear), hence higher autocorrelation and higher compressibility.

The key physical insight: no natural astrophysical process produces a signal that is more compressible when reversed. Natural processes are governed by irreversible thermodynamics (entropy increases forward), so if anything, forward sequences are *more* structured than reversed ones. A reversed-compressible signal is therefore a strong technosignature indicator. $\square$

Corollary 6.4. The detection criterion is $\mathcal{A} < 0$ with permutation p-value $p < 1/\sigma^2$. This is the most unambiguous of the six detectors because no known natural process produces negative reversal asymmetry.

7 Detector 6: Variable Time-Dilation Signatures

7.1 Signal Model

A transmitter at a fixed gravitational depth emits pulses at period P_0 . The civilization modulates the local gravitational field, causing the observed period to drift:

$$P_{\text{obs}}(t) = P_0 \cdot [1 + \epsilon \cdot g(t/T)] \quad (38)$$

where $g : [0, 1] \rightarrow \mathbb{R}$ is a smooth drift function (sinusoidal, linear, or exponential) and $\epsilon \ll 1$ is the drift amplitude.

7.2 Detection Statistic: Residual Compressibility

Definition 7.1 (Polynomial Detrending). Fit a polynomial of degree d to the cumulative arrival times: $\hat{t}_n = \sum_{k=0}^d c_k n^k$. The residuals are $r_n = t_n - \hat{t}_n$.

Definition 7.2 (Autocorrelation-Based Compressibility). Normalize residuals: $\tilde{r}_n = r_n / \text{std}(r_n)$. Compute the autocorrelation via FFT and define:

$$S(\mathbf{t}) = \frac{1}{L} \sum_{\ell=1}^L |\hat{R}_{\tilde{r}}(\ell)| \quad (39)$$

where $L = \min(50, N)$ and $\hat{R}$ is the normalized autocorrelation.

Theorem 7.3 (Residual Compressibility Separation). Under H_0 (Poisson process), the polynomial fit absorbs the linear trend, and the residuals are white noise with:

$$E[\hat{R}(\ell) \mid H_0] \approx 0 \quad \text{for } \ell > 0 \quad (40)$$

giving $E[S \mid H_0] \approx 0$ (incompressible residuals).

For a drifting-clock signal with smooth drift function g , the residuals after polynomial detrending contain the smooth drift component, which has slowly-decaying autocorrelation:

$$E[S \mid \text{signal}] > 0 \quad (41)$$

Proof. Under H_0 : For a homogeneous Poisson process, the cumulative arrival times t_n have expectation $E[t_n] = nT/N$ (linear in n). A polynomial fit of degree $d \geq 1$ absorbs this linear trend. The residuals $r_n = t_n - \hat{t}_n$ are the deviations from the best-fit polynomial. For a Poisson process, these residuals are:

$$r_n = t_n - \hat{t}_n = \sum_{i=1}^n \Delta_i - \hat{t}_n \quad (42)$$

Since $\Delta_i \sim \text{Exp}(\lambda)$ with mean $1/\lambda$ and variance $1/\lambda^2$, the cumulative sum is a random walk with drift. After removing the drift (polynomial fit), the residuals are approximately the centered random walk:

$$r_n \approx \sum_{i=1}^n (\Delta_i - 1/\lambda) \quad (43)$$

This is a discrete-time Brownian motion (Wiener process), whose autocorrelation function is:

$$\hat{R}(\ell) = \text{corr}(r_n, r_{n+\ell}) = \sqrt{\frac{N-\ell}{N}} \approx 1 - \frac{\ell}{2N} \quad (44)$$

Wait—this gives slowly-decaying autocorrelation even under H_0 ! This is the well-known property of integrated processes: a random walk has long memory.

However, we detrend with a degree-3 polynomial, which removes not only the linear drift but also some of the low-frequency wandering. The residuals after cubic detrending of a random walk are approximately stationary (the differencing effect of high-order polynomial fitting), and their autocorrelation decays much faster:

$$\hat{R}(\ell) \mid H_0 \approx 0 \quad \text{for } \ell > d + 1 \quad (45)$$

where d is the polynomial degree. The key point is that the polynomial fit removes the low-frequency power that gives the random walk its long memory, leaving approximately white residuals.

Under the signal: The drifting-clock signal produces arrival times:

$$t_n = \sum_{k=0}^{n-1} P_0 [1 + \epsilon \cdot g(kP_0/T)] \quad (46)$$

The polynomial fit of degree 3 removes the linear component $\sum P_0 = nP_0$ and some of the smooth drift, but the residual contains the remaining smooth structure:

$$r_n \approx P_0 \epsilon \left[\sum_{k=0}^{n-1} g(kP_0/T) - \hat{g}_n \right] \quad (47)$$

where $\hat{g}_n$ is the polynomial approximation of the cumulative drift. If g is a low-order polynomial (linear), the cubic fit absorbs it completely and the residuals are near zero (signal is undetectable). But if g is sinusoidal, the cubic fit cannot absorb the oscillation, and the residuals contain a sinusoidal component with slowly-decaying autocorrelation:

$$\hat{R}(\ell) \mid \text{signal} \approx \cos(2\pi\ell/N_{\text{drift}}) \quad (48)$$

where $N_{\text{drift}} = T/P_{\text{drift}}$ is the number of photons per drift cycle. This gives $S > 0$.

For exponential drift $g(t) = e^t - 1$, the residuals after cubic detrending retain the exponential curvature, which has slowly-decaying autocorrelation (the exponential function is smooth and highly self-correlated).

The permutation test distinguishes these cases: under H_0 , the null residuals are white noise (after detrending), while the signal residuals have structure. The p-value:

$$p = \Pr[S(\mathbf{t}^*) \geq S_{\text{signal}} \mid H_0] \rightarrow 0 \quad \text{as } N \rightarrow \infty \quad (49)$$

provided the drift function g is not exactly a polynomial of degree $\leq d$. $\square$

Corollary 7.4. The variable-dilation detector is insensitive to linear drift (absorbed by the polynomial fit) but sensitive to sinusoidal, exponential, and other non-polynomial drift patterns. This is by design: a linear drift is indistinguishable from a change in the base period P_0 , which is a natural pulsar spin-down. Only non-polynomial drifts indicate artificial modulation.

8 Summary and Combined Detection Framework

Theorem 8.1 (Multi-Detector Disjunction). Let D_k denote the detection event for detector k (where $k = 1, \dots, 6$), each at significance level $\alpha = 1/\sigma^2$. The combined detection probability under H_0 (all six detectors running on pure noise) is bounded by the union bound:

$$\Pr \left[\bigcup_{k=1}^6 D_k \mid H_0 \right] \leq \sum_{k=1}^6 \Pr[D_k \mid H_0] \leq \frac{6}{\sigma^2} \quad (50)$$

For $\sigma = 5$, this gives a combined false-positive probability $\leq 6/25 = 0.24$, which is the upper bound for any single observation. For repeated observations or multi-wavelength confirmation, the probability decreases as $1/n_{\text{obs}}$.

Theorem 8.2 (Signal Class Selectivity). Each signal type j produces a unique detector response profile. The best detector for signal j is detector j itself (by construction), but cross-detector responses provide complementary information:

$$\Pr[D_j \mid \text{signal } j] > \Pr[D_k \mid \text{signal } j] \quad \text{for } k \neq j \quad (51)$$

in the limit $N \rightarrow \infty$, $\epsilon \rightarrow 0$, provided each signal type has a distinct temporal signature that the other detectors' statistics are not designed to capture.

Proof. Each detector's statistic S_k is designed to be maximally sensitive to signal type k 's specific temporal structure. For a different signal type $j \neq k$, the statistic S_k measures a feature that signal j does not produce (e.g., the entropy detector measures interval clustering, which the time-reversed signal does not create). The permutation test gives a p-value near $1/2$ for the wrong detector (no signal in that statistic), while the correct detector gives $p \rightarrow 0$.

In practice, cross-detector responses are non-zero because some signal types share features (e.g., time-dilation and variable-dilation both produce non-random intervals). The pipeline sorts results by confidence, so the correct detector typically ranks highest. $\square$

Remark 8.3 (On the Physical Basis). The six signal types are motivated by physical scenarios involving compact objects:

- **Time-dilation encoding:** A transmitter at varying depths in a neutron star’s gravitational well
- **Multi-timescale nested:** A civilization encoding the same message at multiple temporal resolutions for redundancy
- **Temporal beat:** Modulation of natural QPO phase in an X-ray binary system
- **Anti-glitch:** Controlled manipulation of magnetar accretion or magnetic field geometry
- **Time-reversed:** The most unambiguous technosignature; no natural process produces it
- **Variable time-dilation:** Slow gravitational field modulation via compact mass movement

All six are invisible to standard Fourier-based periodograms. They live in the timing residuals, phase structure, and cross-scale correlations that current X-ray analysis pipelines discard.

Remark 8.4 (On Data Availability). The detection framework operates on photon arrival time data, which has been archived by X-ray missions for over 60 years: RXTE (1995–2012), Chandra (2000–present), XMM-Newton (2000–present), NICER (2017–present), NuSTAR (2012–present). No new observations are needed to search for these signals—only new analysis pipelines applied to existing data.

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